

GENERAL SPECIFICATION G4.10

Air Blowers

CONTENTS

G4.10 Air Blowers	1
G4.10.1 General Requirements	1
G4.10.2 Submittals	7
G4.10.3 Quality Experience	8
G4.10.4 Hazardous Area Requirements	8
G4.10.5 Centrifugal Blower	8
G4.10.6 High Speed Turbo Blower	16
G4.10.7 Positive Displacement Blower	22
G4.10.8 Test and Inspection	25
G4.10.9 Control Requirements	27
G4.10.10 Associated Plant and Equipment	32

G4.10 AIR BLOWERS

G4.10.1 General Requirements

- (a) The blower shall deliver oil free air.
- (b) The blowers shall be constructed in a manner that eliminates vibration occurring to the plant or being transmitted into the building structure.
- (c) The design and all materials shall be selected to suit the installed environment and climatic conditions.
- (d) The blower assembly shall be of a compact modular design, which shall ensure a small footprint area to minimise civil construction costs.
- (e) The blower and all associated equipment shall be supplied as an integrated package with a view to optimising availability, prolonging the period over which the design output is maintained, and prolonging the intervals between service re-conditionings. Appropriate de-rating margins shall be applied as necessary to achieve these ends.
- (f) The choice of blower and blower design shall consider the potential for resonance within the system including those known to occur with multiple blower manifolds.
- (g) The blower silencers, air manifold and the overall air system shall be designed to eliminate the effects of resonant frequencies and vibration over the whole operating range of the blowers.
- (h) Blowers shall be selected from the manufacturer's standard range to meet the total head requirements of the complete system and provide the range of flows specified. They shall have a non-overloading power characteristic and be sized with an allowance for the variations in atmospheric pressure, temperature, and relative humidity as specified in the Particular Specification.
- (i) The blower equipment shall be produced and assembled by the blower manufacturer at a facility owned and operated by the blower manufacturer and under the direct supervision and control of the blower manufacturer.
- (j) The blower manufacturer may provide the blower control panels and adjustable frequency drives, and other components from another manufacturer. However, all of this equipment shall be the responsibility of the blower manufacturer to furnish and coordinate.
- (k) The blowers shall be sized to give maximum operating efficiency at the average conditions. The blowers shall be capable of operating over the stated operating range. Blower power ratings shall be quoted for the minimum, average, and maximum operating conditions.
- (l) The blower shall be capable for operation in parallel with same type of blowers delivering the same flow rates.
- (m) Where cascade control is specified, the blower shall be capable of satisfactory operation in parallel with many other blowers at different flow rates (i.e. only one of the machines is regulated, assuming that the variation on demand is less than the capacity range of the blower).
- (n) The blowers shall be selected and sized to enable continuous increase in air supplied from 45% capacity of one blower to 100% of all duty blowers in parallel at the pressure range required.

- (o) The blowers shall be selected to provide high efficiency operation. Blower Isentropic efficiency shall be greater than 80% across all operation range (with maximum specified inlet air temperature and relative humidity).
- (p) Cooling systems for the blowers shall be self-contained and any heat generated shall be ingested or discharged to the outside of the main building structure.
- (q) Special tools required for maintenance shall be delivered with the blower.
- (r) Lifting
 - (i) The blower and any components weighing over 25 kg that are likely to be removed for maintenance shall either:
 - 1) Incorporate clearly identified, permanent lifting points located to give a safe and balanced lift; or
 - 2) Be designed in such a way that standard lifting accessories (e.g. slings) can be easily attached.
 - (ii) The blower shall be provided with lifting lugs.
- (s) Corrosion Protection
 - (i) All blower components and associated pipework (including valves and fittings) shall be designed and assembled to avoid galvanic corrosion.
- (t) Guarding
 - (i) Guarding shall comply with BS EN ISO 12100 and BS EN ISO 14120 or equivalent;
 - (ii) Guarding of machinery (e.g. belt drives) shall be independent of the acoustic enclosure, if provided;
 - (iii) Blower components and associated pipework that present a burns hazard shall be insulated/guarded as specified;
 - (iv) Components that require regular inspection, cleaning or maintenance shall be readily and safely accessible and where appropriate, easily replaceable; and
 - (v) Components shall be designed to avoid the need for the use of special tools for maintenance. If special tools are required, they shall be provided.
- (u) Support Structure
 - (i) Support structure (baseplate or support frame) and mounting arrangements shall be appropriately designed;
 - (ii) The blower stage and its drive system shall be securely fixed to a rigid support structure;
 - (iii) The support structure material shall be as stated in the tender and suitable protective coating system shall be proposed to S.O. for approval;
 - (iv) The support structure shall be designed to prevent the collection of liquids and debris during operation. The ends of all sections shall be sealed to prevent the ingress of corrosive liquids, after all corrosion resistant coatings have been applied; and
 - (v) If necessary, to prevent excessive levels of vibration being transmitted to the surrounding building or plinth, the support structure shall be provided with anti-vibration mountings.

(v) Air Filters and Silencers

- (i) Air filters on the inlet to blowers shall be of the dry element two stage, high volume, and pleated type and shall have differential pressure gauge to determine when cleaning or element replacement is required;
- (ii) The efficiency of the filter shall be greater than 93% removal for particles 5 microns and larger. The intake silencer shall be located downstream of the filter and it shall be acoustical absorption type with fibre glass packing;
- (iii) The inlet filter/silencer shall be designed for maximum air flow at absolute minimum pressure drop, and connected directly to the inlet of the blower or blower enclosure via a flexible connection; and
- (iv) The filter must have a minimum standard of EU3 in accordance to DIN 24185 or equivalent.

(w) Local Control Panel

- (i) Each blower shall be equipped with a separate Local Control Panel (LCP) which shall comply with requirements for machine PLCs as specified in General Specifications G6 ICA. Each LCP shall include integrated controls that manage stop / start of the blower and its ancillaries. LCP's shall preferably each have a touch-screen HMI;
- (ii) The integral control system shall include all control of blower components as well as comprehensive condition monitoring and associated control interlocks to protect the blowers and associated systems from potential damage. Control philosophy shall be fail safe operation i.e. all controls and instruments shall fail go into fail safe mode in a trip situation. An alarm shall be generated before any trip condition. The controls shall be designed so that the blower cannot operate unless the controls are energised, nor can they operate with any defective controls, unless otherwise specified; and
- (iii) Communication between each Blower LCP and the Blower System Master PLC shall use an Ethernet-based protocol and optical fibre media.

(x) Blower System Master Control Panel

- (i) Each blower system shall be equipped with a Master Control Panel (MCP) which shall comply with requirements for machine PLCs as specified in General Specifications G6 ICA. Requirements for MCP's may differ from those for LCP's;
- (ii) The MCP shall include integrated controls and a touch-screen HMI to manage all the blowers in the system, to meet an overall air flowrate or pressure demand signal generated by the plant control system; and
- (iii) Communication between the MCP and the plant control systems shall use an Ethernet-based protocol and optical fibre media.

(y) Motor

- (i) Each blower unit shall be connected to the motor via gearbox, flexible or direct coupled;

- (ii) All Centrifugal Blower and Positive Displacement Blower motors shall comply with the General Specification-Electrical. All High Speed Turbo Blower motors shall be as specified in section G4.10.6 High Speed Turbo Blower; and
- (iii) All Centrifugal Blower and Positive Displacement Blower motor couplings shall have a power rating of not less than 1.5 times the motor nameplate power when the misalignment is within the coupling manufacturer's tolerance limit.
- (z) Acoustic Enclosures
 - (i) Acoustic enclosure shall be provided for each blower to ensure that noise level at 1 m away from the enclosure is not more than 85 dB(A). Such enclosures are not required only if the generated noise level is lower than 85 dB(A);
 - (ii) The enclosure shall be selected based on its noise reduction performance, maintainability and its ability to withstand the elements;
 - (iii) For major overhaul, the enclosure shall be designed to allow for partial disassembly to allow access to both blower and motor by a lifting device. The enclosure shall be of modular design for easy access and removal;
 - (iv) The enclosure shall incorporate over temperature detection device the output of which shall be suitable for control and monitoring;
 - (v) The enclosure shall be provided with a suitable method of ventilation for blower cooling. If ventilation fans are required, duty / standby ventilation fans shall be provided inside the acoustic enclosure to direct all exhaust air to outside of building; The fans shall be temperature controlled with temperature sensors designed to signal an alarm in the event of the ventilation fan failing due to over temperature;
 - (vi) Sufficient small power and lighting shall be provided inside the enclosure subject to S.O.'s approval;
 - (vii) Filter pressure, loss indicator, and discharge pressure gauge shall be visible from outside the enclosure;
 - (viii) The enclosure shall incorporate hinged doors or lift-off panels to provide safe access to all blower components requiring inspection, cleaning or maintenance. The structural integrity of the enclosure shall be maintained when one or more panels are removed. Doors/panels shall comply with the following requirements, as a minimum:
 - 1) Doors shall open to a minimum angle of 90° and be restrained against wind loads when open;
 - 2) Panels shall be easily removed and manipulated by one person (e.g. fitted with suitable handles) and have a maximum weight of 25 kg;
 - 3) Doors/panels shall be securely held in place when closed/in position and shall be lockable by padlock; and
 - 4) Seals shall be provided around doors/panels to maintain the acoustic performance of the enclosure.

- (ix) Where provided, 'walk-in' type acoustic enclosures shall be provided with all appropriate warning signs/notices;
 - (x) The enclosure shall have an outer skin of steel or thick GRP, and an inner skin of zinc coated perforated steel. The space between the outer and inner skins shall be filled with high density mineral wool, unless otherwise approved by the S.O.;
 - (xi) All routine inspection and servicing shall be achievable without the need to remove the enclosure; and
 - (xii) Enclosure discharge ventilation and pressure relief valve ventilation shall be ducted or vented away from the enclosure so that the high temperature air is dissipated away from the enclosure and any personnel access.
- (aa) Spares
- (i) The Contractor shall ensure that all components/systems that require lubricants, coolants, etc. are provided. All grease lubricated bearings and associated lubrication pipework shall be fully charged with grease.
- (bb) Air Intake
- (i) Air intakes shall be positioned in a manner or/and provided with appurtenances for the prevention of collection of contaminants such as engine exhaust fumes, excessive dust, dirt, for avoidance of rain being drawn in and always be away from any extractor fan discharge points;
 - (ii) The blower inlet air may be admitted to the room or the open space, which houses the blowers, either through an external wall louver ducted directly to the blower filter/silencer or from the space, where the intake duct is connected to the blower filter/silencer. Where air is delivered to processes on a continuous basis, the blowers shall be connected to the intake duct to draw air from outside the building and not from the room itself;
 - (iii) A louvered panel or panels shall be installed at the air inlet point together with filters. The filters shall consist of removable panels with convoluted elements mounted in frames. The elements shall be of the washable type and a spare set shall be provided to facilitate substitution for this purpose;
 - (iv) Unless otherwise specified, the intake ducting shall be in galvanised steel and of adequate proportions to minimise pressure drop and vibration. Where necessary to suppress low frequency noise, a silencer of the multiple aerofoil or 'beam splitter' type shall be installed within the duct close to the inlet point;
 - (v) Where air is drawn by the blowers directly from the room or the space housing the blowers, appropriate provision shall be made to protect equipment and pipework from low ambient temperatures. Where air is drawn from outside the building, heat from the machines must be dissipated effectively at times of high ambient temperatures; and
 - (vi) The room or the space housing the blowers shall be properly ventilated to dissipate the heat generated by the blowers.

- (cc) Exhaust Air
- (i) Unless otherwise specified, all heat produced by the blowers shall be directed to the outside of building; and
 - (ii) Heat produced from the blower's motor oil cooling systems shall be directed to outside of the blower building or be located at outside of the blower building.
- (dd) Vibration Analysis and Vibration Monitoring System (VMS)
- (i) All shaft vibration monitoring system shall be furnished for each blower and motor using acceleration probes, unless otherwise specified;
 - (ii) The system shall include 2 directions X-Y configuration vibration detectors at each bearing of the motor, four (4) probes total per motor;
 - (iii) A full list of the vibration and protection sensors for air blower and motor shall be submitted to S.O. for approval; and
 - (iv) Unless otherwise specified, the temperature sensors and vibration probes shall be provided as per the following as minimum:

Motor per unit		
Winding Temperature	6 numbers	2 per phase; for display (on LCP), 1 per phase.
Vibration	4 numbers	2 numbers for motor drive end and 2 numbers for motor non-drive-end.
Bearing temp	2 numbers	1 number for motor drive end and 1 number for motor non-drive-end.
Drive Shaft per unit		
Vibration	4 numbers	2 numbers for motor drive end and 2 numbers for motor non-drive-end.
Bearing temp	2 numbers	1 number for motor end and 1 number for motor non-drive-end.
Pinion Shaft per unit		
Vibration	4 numbers	2 numbers for motor drive end and 2 number for motor non-drive-end.
Bearing temp	4 numbers	1 number radial bearing for motor non-drive-end and 1 number for motor drive end, 1 number for thrust bearing, 1 number for counter thrust bearing.
Pinion shaft, axial, counter thrust	2 numbers	
Pinion shaft, speed probe	1 numbers	

G4.10.2 Submittals

- (a) Contractor shall provide the following information at Tender stage:
 - (i) A typical general arrangement drawing of the blower indicating the designation, overall dimensions/footprint and typical layout of key components/systems and the following information as minimum:
 - 1) The maximum blower design speed and maximum allowable blower operating speed; and
 - 2) The maximum blower power input.
 - (ii) Typical blower performance curves developed for specified design conditions:
 - 1) Indicating the discharge pressure (kPa), discharge flow (Nm³/h), total wire power (kW) and wire to air efficiency at guarantee points;
 - 2) Furnish performance curves at full speed and a minimum of four lower speeds (or turndown conditions) to indicate specified volume turndown. Include the capacity line above which the unit should be operated to preclude surging. Include the capacity line below which the unit should be operated to preclude run out. Indicate surge and surge margin lines; and
 - 3) Flow ranges (plus or minus for a given operating condition) will not be acceptable.
 - (iii) A component/system schedule or assets list including, for each major component/system, the component/system manufacturer and all relevant technical specifications (including details of materials);
 - (iv) Estimated blower sound level data, for both inlet and radiated conditions, and description of sound control measures required to meet the specified sound levels;
 - (v) Maximum heat dissipation from blower enclosure to the space and any ventilation requirements;
 - (vi) Provide information about internal and external heat exchangers. If external heat exchangers are required, provide remote location installation and mounting requirements;
 - (vii) A lubrication schedule detailing all components/systems requiring lubrication, the method and frequency of lubrication and the type and manufacturer of the lubricants;
 - (viii) A schedule of spares required for 2 years of normal operation (plus associated costs and lead times);
 - (ix) A typical control philosophy, Functional Design Specifications (FDS) and details of PLC's proposed;
 - (x) A typical commissioning plan (including timescales for completion); and

- (xi) With respect to (i), the drawing shall indicate, where appropriate, the minimum clearance distance required for the removal of components (including acoustic enclosures, if required) for refurbishment or replacement with the equipment in situ and details of interfaces with other equipment. The blowers and all required space and clearance for access, maintenance and lifting operations shall fit into the designated locations of the blowers.
- (b) Material certificates for casing, gear, shaft and impeller, as minimum, shall be submitted for S.O.'s approval.

G4.10.3 Quality Experience

- (a) The blower Manufacturer shall have at least ten (10) years' experience in the design, application and supply of air blowers for used water (wastewater) reclamation plant or similar applications. Tender shall submit together with the reference list of not less than ten (10) separate operational installations and shall have at least five (5) installations of equivalent or close to the size of blowers that have been in operation for at least five (5) years as evidence of meeting the experience requirement.
- (b) The manufacture must be ISO 9001:2015 certified to assure the conformance to the highest quality standards of the industry.

G4.10.4 Hazardous Area Requirements

- (a) All equipment selected for use in hazardous areas shall have undergone an appropriate conformity assessment procedure (CAP) to demonstrate compliance with the essential health and safety requirements of European Directive 94/9/EC (ATEX 95) or equivalent.

G4.10.5 Centrifugal Blower

- (a) General
 - (i) Unless otherwise specified, the centrifugal blowers shall be single stage, integrally geared type designed for energy optimisation, high operating efficiency and low maintenance costs. The running costs for the design life of the plant shall be evaluated to achieve maximum efficiency;
 - (ii) The centrifugal blower shall be able to provide variable flow rate, controlled by a combination of inlet and outlet guide vanes with fixed speed motors;
 - (iii) A fully automated blower control system shall be included as part of the blower installation to monitor critical parameters of the blower and to control performance;
 - (iv) The blowers shall be supplied, as applicable, with the following features:
 - 1) A blow-off/unloading valve, silencer and failure detection system;
 - 2) Pressure relief valve set to 10% above system operating pressure, and capable of 100 % blower discharge flow fitted with a pressure accumulation not more than 10% of set pressure;
 - 3) In built self-diagnostics together with real time condition monitoring;
 - 4) Inlet process air filter;

- 5) A surge protection system;
- 6) Inlet filter differential pressure indicator local to the filter. An electronic alarm signal shall be integral with the controls;
- 7) Inlet silencer;
- 8) Discharge silencer and compensation bellows in pipeline;
- 9) Inlet and outlet silencers of the reactive, absorptive or packed type shall be selected with consideration of the Pitch Line Velocity (PLV), maximum noise reduction and minimum pressure loss through the silencer;
- 10) Suitably sized non-return check valve on blower discharge;
- 11) An isolating valve;
- 12) An outlet pressure gauge with isolating valve. The gauge full scale deflection shall be between 1.5 and 2.0 times the maximum outlet pressure;
- 13) An inlet air temperature sensor, with high level switch (to detect exceedingly high temperature such as more than 40 deg. C. above the ambient temperature or as recommended by the Manufacturer);
- 14) A vibration sensor with high level switch;
- 15) High discharge air temperature alarm; and
- 16) Diffuser according to blower type and application.

With respect to item 1) above, the blow-off/unloading valve shall discharge to a safe position, away from the blower inlet.

With respect to item 15) above, the costs associated with the requirement for any additional auxilliary cooling sub-system to cope with any normal or extreme operating conditions that shall be encountered shall be deemed to be included in the tender sum.

- (v) Operation of several centrifugal blowers on line shall be possible without special requirements. The condition of several units running in parallel with different air flow settings shall be met and the safety margin between discharge pressure and surge pressure shall be maintained at the same value for each machine operating in parallel;
- (vi) Starting or stopping of any individual blower shall be possible without altering the airflow of the plant;
- (vii) A surge control must be installed and designed to detect the incidence of pneumatic surge;
- (viii) The safety margin between the maximum outlet pressure required and the surge pressure shall be a minimum of 3.5 % of the maximum outlet pressure required;

- (ix) The blowers shall have a performance range to enable satisfactory flow diversion and transition between blowers in parallel operation. This shall be achieved by cascade control to provide the most efficient means of operation. Any alternatives shall be proposed and submitted to the S.O. for approval. Each blower shall be capable of achieving the capacity turndown from 100 to 45 percent under all operating conditions;
 - (x) All losses for silencers, check valves and the drive system shall be included in the total power figure stated in the tender and the blower output shall be rated to take such losses into account; and
 - (xi) Unless otherwise specified, the blower shall not exceed the maximum vibration levels as specified in BS ISO 10816.
- (b) Mechanical Construction
- (i) For the single stage centrifugal blower, it shall be integrally geared type with integral electric drive motor and gearbox. The volumetric air output shall be controlled over the required range by mechanical means such as variable inlet guide vanes and variable discharge diffuser vanes;
 - (ii) All parts shall be made and assembled from one manufacturer to ensure the quality of the product;
 - (iii) The casing shall be made of grey cast iron to BS EN 1563 Grade 420-12 SG or DIN 1693 GGG40 as minimum or equivalent;
 - (iv) The blower unit shall be horizontally mounted on the base plate of the packaged unit and shall be connected to the conical diffuser by clamps with rubber sealing elements;
 - (v) The conical diffuser and the blow off valve shall be mounted directly integrated into the acoustic hood and shall be sufficiently and safely supported by a steel supporting structure to protect the blower unit against external load;
 - (vi) The conical diffusers shall be designed, tested, and manufactured according to the pressure equipment directive PED 97/23/EG or equivalent standard; and
 - (vii) The check valve and the compensator with flange connection shall be provided. Check valve shall be provided on the pressure side of the blower and its material shall be stainless steel.
- (c) Impeller and Shaft Seals
- (i) The design of the impeller shall incorporate a backward curved vane profile for inlet guide vane control and radial impeller for variable diffuser control;
 - (ii) The impeller shall be manufactured from forged aluminium alloy or high grade alloy steel. Fabricated steel or aluminium impellers shall not be used;
 - (iii) Machining of entities made of Aluminium alloy or high-grade alloy steel shall be done as a whole unit, not welded, casted or riveted;
 - (iv) The design shall be such that vibration or flexing on the shaft does not occur within the operating speed range of the blower;

- (v) The impeller shall be attached to the shaft by shrink fit and locknut arrangement. The shrink fit construction shall ensure sufficient attachment and centring of the impeller at all operating conditions. The impeller shall be dynamically balanced;
 - (vi) The design of the rotating element shall ensure that the first critical speed does not occur within +/- 20 % of the operating speed range of the blower;
 - (vii) Any responsive lateral critical speed of the rotating assembly shall be at least fifteen (15) percent from the normal operating speed;
 - (viii) Any torsional resonances of the package shall be at least ten (10) percent from the normal operating speed; and
 - (ix) All shafting shall conform to "Design and Selection of Components for Enclosed Gear Drives" (AGMA 6001-D97).
- (d) Volute Casing
- (i) Inner Volute casings shall be made of grey cast iron and the outer volute shall be ductile cast iron. Discharge flanges shall be drilled to BS EN 1092 PN10 rating as a minimum. All joints of the casing shall be machined;
 - (ii) Seals shall be of the non-contact, multi-point labyrinth type and be readily replaceable. The high-speed air and oil seals shall be located in a vented chamber;
 - (iii) Venting and drainage facilities shall be provided at the highest and lowest points on the casing;
 - (iv) The volute casing shall be mounted on the gear case flange. All mating surfaces shall be accurately machined and be provided with deep registers, dowels and spigots where necessary to ensure correct alignment;
 - (v) The volute casing outlet shall be provided with a flexible expansion joint to accommodate vibration and/or expansion or contraction of the outlet pipework. If necessary, the expansion joint shall be protected from distortion during service by tie-rods; and
 - (vi) The volute casing outlet shall be provided with an outlet cone to maximise the recovery of velocity head.
- (e) Variable Guide Vanes and Variable Diffuser Vanes
- (i) An inlet variable guide vane assembly and an outlet variable diffuser vane system shall be mounted integral with each blower. The inlet guide vanes shall be made in an aerodynamic, streamlined design in cross section and located radially;
 - (ii) The inlet guide vanes and outlet diffuser vanes shall be able to regulate the Isentropic Head (pressure and temperature) and the flow to optimise the blower efficiency;
 - (iii) The regulation should be fully automated by means of a program for continuous optimisation of blower efficiency with respect to changes in inlet temperature, differential pressure and the required flow. The regulation program must be individually designed for each blower and based on the measured result from the performance test for each blower;

- (iv) Variable diffuser vanes shall be provided for capacity control and designed to obtain high diffuser efficiency in the entire regulating range;
 - (v) The purpose of a combined inlet guide vane and variable diffuser vane system shall be to facilitate the turn down of each blower from 100 to 45 % air flow rate capacity. Moreover, efficiency shall be maximised over the entire turn down range, especially when the blower is operating under off design conditions (lower temperature and/or pressure); and
 - (vi) Full details of any variable inlet guide vane and/or variable diffuser vane systems supplied with the blower for regulating blower performance shall be provided with the tender.
- (f) Blower Efficiency
- (i) The blower shall achieve and maintain a high volumetric efficiency. Volumetric efficiency shall not be allowed to fall below a level where it adversely affects air temperature rise and overall blower efficiency.
- (g) Impeller Shaft Seals
- (i) The shaft seals shall be of a non-contact, multi-point, labyrinth type and operated dry;
 - (ii) The impeller shaft shall be provided with seals to prevent the ingress of oil and contaminants into the volute casing and avoid contamination of the gearbox oil;
 - (iii) Shaft sealing arrangements shall ensure oil free air delivery;
 - (iv) The shaft sealing arrangement shall ensure a process air quality of Class 0 in accordance with ISO 8573-1;
 - (v) Rotating components and shaft seals shall withstand the pressures, temperatures, and corrosive/erosive properties of the air;
 - (vi) A vented space between air and oil seals shall be provided;
 - (vii) Any leakage shall be minimised by having small clearances between female and male parts. The female part shall be made of aluminium or bronze to avoid damage to the shaft in the event of a seal rub; and
 - (viii) Uniform shaft diameter without multiple slinger rings in each sealing area will not be acceptable.
- (h) Gears
- (i) The rotor shall be driven by a step-up gearbox. Speed increasing gear shall be of the helical type. The pinion shall be integral with the rotor shaft. The gears and shafts shall be made of case hardened alloy steel forgings with gear teeth precision ground;
 - (ii) All gears shall be rated in accordance with “Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth” (AGMA 2101-C95) or equivalent;
 - (iii) The gears shall be manufactured to a minimum AGMA quality number of twelve (12) per “Gear Classification and Inspection Handbook” AGMA 2000-A88 or equivalent;

- (iv) The gearbox shall be of ample size and rated to transmit the maximum torque and power input requirements to the blower under all operating conditions;
 - (v) The gearbox shall incorporate a pressurised oil lubrication system to lubricate the gears and bearings. All components shall be installed integral with the blower base plate and arranged to permit ease of accessibility for operation, maintenance, inspection, and cleaning. The system shall incorporate the following items, as a minimum:
 - 1) An oil reservoir;
 - 2) A main oil pump (mechanically driven) with suction strainer;
 - 3) An auxiliary oil pump (electrically driven) with suction strainer;
 - 4) An oil cooler;
 - 5) An oil filter with visual condition indicator;
 - 6) All necessary pressure relief valves, check valves, fittings and pipework etc. to ensure effective, fail-safe operation;
 - 7) All necessary instrumentation and protective devices to ensure effective, fail-safe operation (including, as a minimum, oil temperature and pressure and bearing temperature monitoring sensors); and
 - 8) Readily accessible oil drain and fill/vent points to allow oil removal/replacement in-situ without excessive spillage and without disturbing blower mountings and/or ancillary equipment. An effective oil mist vent filter shall be provided.
 - (vi) The auxiliary oil pump shall be provided to ensure effective gearbox lubrication during blower start-up and run-down. It shall also act as a standby unit if the main oil pump fails.
- (i) Bearings
- (i) Hydrodynamic, oil pressure lubricated bearings shall be required with sufficient oil film thickness under all operating conditions;
 - (ii) All bearings shall be of bronze construction and having a minimum L-10 life of 100,000 hours;
 - (iii) Slow speed shaft radial bearings shall be cylindrical, journal type (hydrodynamic bearing);
 - (iv) Fast shaft radial bearings shall be multi-segment and babbitted, designed to suppress hydrodynamic instabilities and provide sufficient dampening to limit rotor vibrations. Fast shaft thrust bearings shall be multiple segment, tapered land type and designed for thrust in both directions. Material bronze alloy; and
 - (v) Rolling element (i.e. ball bearings) are not allowed to be used for any bearing in the gearbox when main motor nominal power is equal or higher than 250 kW.

- (j) Surge Protection (if applicable)
- (i) An electrically actuated blow-off valve venting directly to atmosphere shall prevent surge conditions arising in the process air pipework. The valve shall be opened during blower start up and when the air flow is less than that at surge point; and
 - (ii) An automatic control system shall be included to monitor the air flow/pressure and modulate the blow off valve as required. The control system shall automatically open the blow off valve and shut down the air blowers on detection of a surge condition.
- (k) Motor
- (i) The motor shall comply with the relevant sections of the General Specification on motors;
 - (ii) Flexible shaft couplings and guard shall be replaceable without disturbing the blower stage and/or its drive (i.e. spacer-type couplings shall be fitted). Couplings shall be designed with a service factor of 1.5 as minimum (based on the maximum blower power input); and
 - (iii) Where the V-belt drive is required, the V-belt transmission shall incorporate an automatic belt-tensioning device. V-belt drives shall be designed with a service factor of 1.5 as minimum (based on the maximum blower power input) and shall be subject to S.O.'s approval.
- (l) Base Frame
- (i) The motor, blower and gearbox (if specified) assembly shall be mounted on a base frame irrespective whether the motor/blower arrangement is vertical or horizontal. The base frame shall be equipped with anti-vibration mountings to prevent the transmission of vibration to the support structure, silencer, filters or connected pipework. Details of all static and dynamic loads transmitted to the foundations shall be included in the Contractor's design procedures; and
 - (ii) The mounting mounts shall be St.SIS 1312 or better for damping natural rubber hardness 60.

(m) Service Factors

The blower, incl. motor, shall as a minimum fulfil following service factors:

Gear	≥ 1.8 according to DIN 3990 Part 21
Bearing	≥ Minimum life time of 100.000 hours at any axis
Coupling	≥ 1.5

(n) Instrumentation and Control Signals

- (i) The minimum instrumentation shall be included as following:
 - 1) Inlet filter differential pressure gauge and transmitter;
 - 2) Inlet air high temperature sensor/switch;
 - 3) Surge indication switch (surge monitor);

- 4) Oil high temperature sensor/switch;
 - 5) Oil temperature gauge;
 - 6) Oil low pressure sensor/switch;
 - 7) Oil pressure gauge;
 - 8) Oil filter differential pressure indicator;
 - 9) Self-acting thermostatic valve for oil cooler;
 - 10) Acoustic enclosure high temperature sensor;
 - 11) Blower outlet air temperature transmitter;
 - 12) Vibration sensors on motor and gearbox;
 - 13) Variable diffuser position transmitter and indicator; and
 - 14) Inlet guide vane position transmitter and indicator.
- (ii) Minimum controls signals which shall be provided as following:
- 1) Local/Auto;
 - 2) Running / Stopped;
 - 3) Oil pump running;
 - 4) Blow off valve open/closed;
 - 5) Diffuser position;
 - 6) Inlet guide vane position;
 - 7) Current blower flowrate output (calculated);
 - 8) Blower motor tripped/healthy;
 - 9) Blower motor temperature normal/high;
 - 10) Acoustic Enclosure Temp high;
 - 11) Air Inlet temp high;
 - 12) Air Outlet temp reading and high alarm;
 - 13) Vibration warnings;
 - 14) Vibration tripped/healthy;
 - 15) Oil pressure normal/high;
 - 16) Surge event; and
 - 17) Recirculation event.
- (iii) The minimum requirement for blower control shall be as follows:
- 1) Variable inlet guide vane (VIGV) and variable diffuser vane (VDV) systems shall be automatically adjusted according to the Control System settings;
 - 2) The blow-off/unloading valve shall automatically open during blower starting and shut-down;

- 3) If a number of blowers are operating in parallel, starting and stopping of any individual blower shall be possible without altering the overall flow rate of the plant; and
- 4) The blower shall automatically shut-down (or starting shall be inhibited) in any case that may cause damage to the blower or other plant. These conditions shall include but not be limited to the following fault/alarm conditions:
 - Low low - gearbox oil pressure;
 - High high - gearbox oil temperature;
 - High high - gearbox bearing temperature;
 - High high - inlet air temperature (reverse flow);
 - Blow-off/unloading valve fault;
 - Surge conditions;
 - High High outlet air temperature;
 - High High acoustic enclosure internal temperature; and
 - High High blower vibration.
- 5) Re-starting of the blower shall not be possible unless the fault/alarm is manually reset at the LCP.

G4.10.6 High Speed Turbo Blower

(a) General

- (i) Turbo blowers shall be complete pre-packaged units, designed for energy optimisation, high operating efficiency and low maintenance costs. The running costs for the design life of the plant shall be evaluated to achieve maximum efficiency.
- (ii) Turbo blowers shall be supplied, as applicable, with the following features as minimum:
 - 1) A blow-off/unloading valve, silencer and failure detection system;
 - 2) In built self-diagnostics together with real time condition monitoring;
 - 3) Inlet process air filter;
 - 4) A surge protection system;
 - 5) Inlet filter differential pressure indicator local to the filter. An electronic alarm signal shall be integral with the controls;
 - 6) Inlet silencer;
 - 7) Discharge silencer and compensation bellows in pipeline;
 - 8) Suitably sized non-return check valve on blower discharge;
 - 9) An isolating valve;

- 10) An outlet pressure gauge with isolating valve. The gauge full scale deflection shall be between 1.5 and 2.0 times the maximum outlet pressure;
- 11) An outlet temperature sensor, with high level switch (to detect reverse flow);
- 12) A vibration sensor with high level switch;
- 13) Blow off (Bypass) valve and silencer;
- 14) High discharge air temperature alarm;
- 15) Diffuser according to blower type and application; and
- 16) Bearing failure system and monitoring.

With respect to item (1), the blow-off/unloading valve shall discharge to a safe position, away from the blower inlet.

(b) Design Selection

- (i) Turbo blower shall be suitable for its intended application as specified. The volumetric performance of the blower (i.e. its flow rate and pressure characteristics) shall meet the requirement as specified in the Particular Specification;
- (ii) The material for all components shall be compatible with the environment as specified in the Particular Specification;
- (iii) Operation of several blowers on line shall be achieved without special requirements. The condition of several units running in parallel with different air flow settings shall be met and the safety margin between discharge pressure and surge pressure shall be maintained at the same value for each machine operating in parallel;
- (iv) Starting or stopping of any individual blower shall be achieved without altering the airflow of the plant;
- (v) Blowers of a design requiring synchronisation of airflow in order that two or more may operate in parallel will not be acceptable;
- (vi) A surge control must be designed and installed to detect the incidence of pneumatic surge;
- (vii) The safety margin between the maximum outlet pressure required and the surge pressure shall be a minimum of 3.5 % of the maximum outlet pressure required;
- (viii) The blowers shall have a performance range to enable satisfactory flow diversion and transition between blowers in parallel operation. Each blower shall be capable of achieving the capacity turndown from 100 to 45 percent under all operating conditions;
- (ix) All losses for silencers, check valves and the drive system shall be included in the total power figure stated in the tender and the blower output shall be rated to take such losses into account;
- (x) Unless otherwise specified, vibration levels shall not exceed 1 mm/s (rms);

- (xi) Turbo blowers shall be suited for the continuous air volume control by control of the impeller speed and suitable for constant and sliding pressure control or direct oxygen control in the basin by connecting an oxygen probe;
 - (xii) A fully automated blower control system shall be included as part of the blower installation to monitor critical parameters of the blower and to control performance. The air blowers shall be fitted with an inlet filter to remove debris and dust particles;
 - (xiii) The air blower intake shall be assessed for risks of corrosive or explosive gases that may exist on site and the air blower exhaust output shall also be assessed for potential risks; and
 - (xiv) The blower shall be oil free and shall be air cooled or include an internal closed-loop glycol cooling system.
- (c) Mechanical Construction
- (i) The blower unit shall include impeller and spiral casing which are directly flanged to the motor housing. The blower impeller shall be mounted directly on the drive shaft of the motor;
 - (ii) The blower unit shall be horizontally mounted on the base plate of the packaged unit and shall be connected to the conical diffuser by clamps with rubber sealing elements;
 - (iii) The conical diffuser and the blow off valve shall be mounted directly integrated into the acoustic hood and shall be sufficiently and safely supported by a steel supporting structure to protect the blower unit against external load;
 - (iv) The conical diffusers shall be designed, tested, and manufactured according to the pressure equipment directive PED 97/23/EG or equivalent standard; and
 - (v) The check valve and the compensator with flange connection shall be provided. Check valve shall be provided on the discharge of the blower and its material shall be stainless steel.
- (d) Impeller and Shaft Seals
- (i) Shafts and impellers shall be statically and dynamically balanced;
 - (ii) Shaft, impeller and motor housing shall be made of stainless steel; and
 - (iii) Impeller shall be machined from a solid forging of aluminium alloy, Type 7075 or made of high strength stainless steel casting to ASTM-A564 grade 630 or equivalent as minimum.
- (e) Unless otherwise specified, variable frequency drive shall be provided for continuous control of the process air.
- (f) Heat Rejection and Cooling
- (i) Blowers shall not allow heat caused by motor or electrical cooling to be exhausted into blower room. Blowers that require any external cooling devices such as external heat exchangers shall be supplied with all necessary appurtenances by the blower manufacturer.

- (ii) Blowers shall be provided with either internal closed-loop glycol cooling unit, with all waste heat ingested into process air or with the ability to divert waste heat from the blower, motor and VFD outside the Blower Facility. Cooling system shall include heat exchangers, fans, pumps and reservoirs and shall be supplied with all necessary appurtenances by the blower manufacturer. Cooling units shall be powered and controlled from the blower control panel.
- (g) Connections and Peripherals
 - (i) The intake air shall be via either front side of the machine or direct connection via the roof of the machine;
 - (ii) The operation of the Blow-Off-Valve opening shall be under the acoustic hood or connected to a piping system through an exhaust at the roof of the acoustic hood;
 - (iii) The exhaust air of the cooling motor shall be via bottom of the acoustic hood, or via roof of the acoustic hood or by connecting a ducted exhaust system as specified; and
 - (iv) Bellow type expansion joint shall be with guiding pipe and control rod. The metal bellow, guiding pipe and welding neck flange shall be made of stainless steel.
- (h) Motor
 - (i) Each blower shall be supplied with a Permanent Magnet Synchronous Motor (PMSM) high speed motor that has no physical connection between stator and shaft, therefore eliminating brushes, slip rings or break resistors. The PMSM shall be combined with a Sine-Wave Filter and Input Line Reactor to maintain cool motor operation and constant motor efficiency with motor turndown. Induction or Brushless DC motors shall not be acceptable.
 - (ii) The motors shall be able to start under starting condition required. Blower manufacturer shall be responsible for coordinating the starting torque requirement of the blower and the motor with VFD to ensure proper operation of the system.
 - (iii) The motor shall have a guaranteed efficiency of 95% at full load.
 - (iv) Insulation Class of the motor shall be Class H. Internal thermocouple shall be provided for motor temperature protection with triple redundancy.
 - (v) If the drive motor is cooled by a separate cooling air blower, it shall consist of permanent-magnetic drive motor and the blower unit shall be directly flanged. Continuous motor shaft speed shall be regulated via variable frequency drive to get the best efficiency of the performance and the details shall be submitted to the S.O. for approval; and
 - (vi) Integrated piping silencer for discharge air of cooling motor shall be provided. Absorption discharge silencer shall be stainless steel unless otherwise specified.
- (i) Variable Frequency Drives
 - (i) Variable frequency drive (VFD) shall be pulse width modulation design, with all necessary components provide a complete with all necessary components and functioning blower system capable of meeting the design requirement.

- (ii) VFD manufacturer shall have a service and support facility in Singapore.
 - (iii) VFDs shall be integrally mounted in the blower enclosure.
 - (iv) VFDs shall have a sinusoidal filter consisting of an inductor and capacitor.
 - (v) VFD and associated harmonic mitigation equipment shall be integrally mounted within blower enclosure or in a standalone enclosure.
 - (vi) If necessary, VFD shall be provided with internally mounted input line reactors as required to meet the specified harmonic limits.
 - (vii) VFD provided for each blower shall have a components and design strategy to mitigate the impacts of heat and stress on PMSM that decreases the motor life. Each VFD shall have an integrated user interface that include field bus connection and support software. Provide control of VFD via blower LCP touch screen.
 - (viii) Harmonic distortion limit shall be in compliance with General Specification G5.40- Low Voltage Frequency Drives.
- (j) Bearing
 - (i) The blowers shall be of the air foil bearing type (smooth or bump) or leaf style air bearing and shall not require oils or lubricants for adequate operation.
 - (ii) The bearings shall be designed to a minimum of 20,000 starts or a minimum of 10 years between replacements.
 - (iii) The blowers shall withstand up to 5,000 shutdowns under full power failure without damage to the bearings.
- (k) Blow-OFF-Valve
 - (i) The valve shall be cone seal by rubber membrane, spring-loaded, pneumatically operated and mounted at the conical diffuser with piping silencer integrated at the valve outlet; and
 - (ii) The valve shall be electronically controlled by solenoid valve to start and stop the blower against external back pressure.
- (l) Cabinet-mounted AC Unit
 - (i) Provide a cabinet-mounted AC unit for each blower unit. The AC unit shall be designed to maintain the blower electronics compartment at 30 degrees C with ambient room air at 40 degrees C. The air conditioning unit shall be powered from and controlled by the blower package.
- (m) Blower Efficiency
 - (i) The blower shall achieve and maintain a high volumetric efficiency. Volumetric efficiency shall not be allowed to fall below a level where it adversely affects air temperature rise and overall blower efficiency. The blower speed shall not be turned down to the substantial detriment of volumetric efficiency or cooling.

- (n) Instrumentation and Control Signals
- (i) The minimum instrumentation shall be included as following:
- 1) Inlet filter differential pressure gauge and transmitter;
 - 2) Inlet air high temperature sensor / switch;
 - 3) Surge indication switch (surge monitor);
 - 4) Blower thermal protection high temperature sensor; and
 - 5) Blower outlet air temperature transmitter.
- (ii) Minimum controls signals which shall be provided as following:
- 1) Local/Auto;
 - 2) Running / Stopped;
 - 3) Blow off valve open/closed;
 - 4) Current blower flowrate output (calculated);
 - 5) Blower motor tripped/healthy;
 - 6) Blower motor temperature normal/high;
 - 7) Blower protection Temp high;
 - 8) Air Inlet temp high;
 - 9) Air Outlet temp reading and high alarm;
 - 10) Vibration warnings;
 - 11) Vibration tripped/healthy;
 - 12) Surge event; and
 - 13) Recirculation event.
- (iii) The minimum requirement for the control of the blower as following:
- 1) Variable frequency drive system shall be automatically adjusted according to the Control System settings;
 - 2) The blow-off/unloading valve shall automatically open during blower starting and shut-down;
 - 3) If a number of blowers are operating in parallel, starting and stopping of any individual blower shall be possible without altering the overall flow rate of the plant;
 - 4) The blower shall automatically shut-down (or starting shall be inhibited) in any case that may cause damage to the blower or other plant. These conditions shall include but not be limited to the following fault/alarm conditions:
 - High high - inlet air temperature (reverse flow);
 - Blow-off/unloading valve fault;
 - Surge conditions;
 - High High outlet air temperature;

- High High acoustic enclosure internal temperature;
- High High blower vibration; and
- High inlet filter differential pressure (if appropriate).

Re-starting of the blower shall not be possible unless the fault/alarm is manually reset at the LCP.

G4.10.7 Positive Displacement Blower

(a) General

- (i) The blower shall be provided with the following items, as a minimum:
 - 1) A blow-off/unloading valve and silencer;
 - 2) A pressure relief valve (PRV), set at 110 % of the maximum operating pressure and rated to pass the maximum blower flow rate;
 - 3) A non-return check valve;
 - 4) An outlet pressure gauge. The gauge full scale deflection shall be between 1.5 and 2.0 times the maximum outlet pressure;
 - 5) Flexible connections to the blower stage inlet and outlet, to prevent transfer of vibration; and
 - 6) Inlet and outlet silencers.
- (ii) With respect to item (1), the blow-off/unloading valve shall discharge to a safe position, away from the blower inlet;
- (iii) With respect to item (2), the PRV shall discharge outside the acoustic enclosure (if provided). Full details of the PRV shall be provided with the tender;
- (iv) With respect to item (5), the inlet and outlet connection details (pipework sizes and flange details) required/provided shall be as stated in the tender submission; and
- (v) With respect to item (6), outlet silencers shall be of the reactive type.

(b) Rotary Lobe Blower

- (i) The Rotary lobe blower shall incorporate twin, lobed rotors, rotating in opposite directions in the blower casing. Rotor shaft rotation shall be synchronised by timing gears to prevent contact between the rotors.

(c) Rotary Screw Blowers

- (i) The blower shall incorporate a pair of asymmetric, screw type rotors, rotating in opposite directions in the blower casing and resulting in internal compression within the blower. Rotor shaft rotation shall be synchronised by timing gears to prevent contact between the rotors.

(d) Design Selection

- (i) The volumetric performance of the blower, including turndown, (i.e. its flow rate and pressure characteristics) shall meet the requirement as specified in the Particular Specification;

- (ii) Blowers with a motor rating up to 15 kW shall be limited to 70 % of the blowers' maximum design speed. Blowers with a motor rating of 15 kW and above shall not exceed 2800 rpm or 70 % of the blowers maximum design speed whichever is lower;
 - (iii) Operating pressure should be no greater than 85 % of the blower's maximum continuous pressure rating;
 - (iv) The blower shall be capable of satisfactory operation in parallel with a number of other blowers;
 - (v) The blower shall achieve and maintain a high volumetric efficiency. Volumetric efficiency should not be allowed to fall to a level where it adversely affects air temperature rise and overall blower efficiency. Variable speed blowers shall not be turned down to the substantial detriment of volumetric efficiency or cooling;
 - (vi) Where variable output is required, this shall be achieved by electric squirrel cage motor and static inverter. The blower, motor, inverter and cabling installation shall be designed as a complete system;
 - (vii) All losses for silencers, check valves and the drive system shall be included in the total power figure stated and the blower output shall be rated to take such losses into account; and
 - (viii) The blower stage and drive system shall be effectively isolated from its support frame and acoustic enclosure to prevent transmission of noise/vibration to the surrounding environment.
- (e) Blower Casing
- (i) The casing shall be fitted with a front cover plate which can be easily removed to give access to the rotating components;
 - (ii) The front cover plate shall be of a wear resistant material and fully compatible with the casing material. When the weight of the cover exceeds 25 kg, provision shall be made for its support during removal; and
 - (iii) Shaft seal arrangements shall ensure an oil free air delivery.
- (f) Rotors and Rotor Shaft
- (i) Rotating components and seals shall withstand the pressures, temperatures and corrosive properties of the blower air;
 - (ii) Rotors and shaft shall be suitably marked to ensure correct alignment and synchronisation on re-assembly;
 - (iii) The rotor shafts shall be provided with seals to prevent the ingress of oil into the casing and to avoid contamination of the timing gearbox oil;
 - (iv) The sealing arrangements shall be as submitted and approved by the S.O.;
 - (v) The sealing arrangements shall ensure a process air quality of Class 0, in accordance with ISO 8573-1;
 - (vi) All rotating assemblies shall be statically and dynamically balanced;
 - (vii) All critical speeds shall be at least 20 % higher than maximum operating speed;

- (viii) The rotating assemblies shall be supported on roller bearings having a minimum L-10 life of 100,000 hours;
- (ix) The rotors shall be dynamically balanced and not exceed the maximum vibration levels as specified in BS ISO 20816-1 and relevant parts of BS ISO 10816;
- (x) For rotary lobe blower the following requirements shall be complied with:
 - 1) The rotors and rotor shafts shall not be machined from a single forging/casting, the method of fixing the rotors to the shafts together with the information of the rotor and shaft materials shall be submitted to and approved by the S.O.; and
 - 2) Lobes shall be positively located on their shafts.
- (xi) For rotary screw blower the following shall apply:
 - 1) The rotors and rotor shafts shall be machined from a single forging/casting. The rotor geometry and the rotor/ shaft material shall be submitted and approved by the S.O..
- (g) Gearbox
 - (i) The rotors shall be driven by a permanent mesh gearbox. The gearbox shall be integral with the blower casing;
 - (ii) Cooling, if required, shall be by air. Lubrication shall be self-contained;
 - (iii) The timing gear life and service factor shall be stated in the tender submission;
 - (iv) The timing gears shall be oil lubricated and method of lubrication shall be stated in tender submission;
 - (v) The timing gear lubrication system for rotary screw blower shall incorporate readily accessible oil filling and drain points to allow oil removal/replacement in-situ without excessive spillage and without disturbing the blower stage mountings and/or ancillary equipment. Oil level windows/sight glasses shall be provided; and
 - (vi) The gear case for rotary lobe blower shall incorporate readily accessible oil filling and drain points to allow oil removal/replacement in-situ without excessive spillage and without disturbing blower stage mountings and/or ancillary equipment. An oil level window/sight glass shall be provided.
- (h) Bearings and Bearing Lubrication
 - (i) All bearings shall be rated for a minimum L-10 life (or equivalent) of 40,000 hours for rotary lobe blower unless otherwise specified;
 - (ii) All bearings shall be rated for minimum L-10 life (or equivalent) of 60,000 hours for rotary screw blower, unless otherwise specified;
 - (iii) The drive end and non-drive-end bearing details (i.e. type, manufacturer, designation, etc.) and method of lubrication shall be submitted to and approved by the S.O.;

- (iv) Automatic grease dispensers, if provided for rotary lobe blower, shall provide a clear indication of grease contents. The minimum capacity/life of the grease dispensers shall be as stated in the tender;
 - (v) If oil-lubricated bearings are provided for rotary lobe blower, bearing housings shall incorporate readily accessible oil filling and drain plugs to allow oil removal/replacement in-situ without excessive spillage and without disturbing blower stage mountings and/or ancillary equipment. Oil level windows/sight glasses shall be provided; and
 - (vi) Oil lubrication systems' if provided for rotary screw blower, it shall incorporate readily accessible oil filling and drain points to allow oil removal/replacement in-situ without excessive spillage and without disturbing the blower stage mountings and/or ancillary equipment. Oil level windows/sight glasses shall be provided.
- (i) Motor
 - (i) Where pulleys and belts are used the motor shall be fitted to slide rails to allow belt tensioning and removal; and
 - (ii) The proposal of the suitable method of blower cooling shall submitted to and approved by the S.O..
 - (j) Base Frame
 - (i) The blower, gearbox and motor shall be mounted on a rigid base frame. Components shall be located by fitted dowels to ensure accurate axial alignment. Anti-vibration mountings shall prevent the transmission of vibration to the support structure, silencers, filters or connected pipework. The base plate shall be surface protected; and
 - (ii) Details of all static and dynamic loads transmitted to the foundations shall be provided.
 - (k) Instrumentation
 - (i) The blower shall be provided with the following items as minimum:
 - 1) A high outlet air pressure sensor; and
 - 2) A high outlet air temperature sensor.
 - (ii) Where a unit is to be used in parallel with other machines, or will need to start against a pressurised discharge, unloading valves for starting shall be provided. Valve position switches and recirculation air temperature switches shall be provided for control and monitoring.

G4.10.8 Test and Inspection

- (a) The blower shall be factory performance tested in accordance with the relevant standards as follows:
 - (i) For Centrifugal Blowers:
 - 1) BS ISO 5389 – turbo compressors –performance test code.
 - (ii) For High Speed Turbo (HST) Blowers:
 - 1) ASME PTC10 Performance Test Code on Compressors and Exhausters.

(iii) For Positive Displacement Blowers:

- 1) BS 1571-2 – Specification for testing Positive displacement compressors and exhausters.
- (b) Submit factory test plan outlining all required testing activities within this section for S.O.'s approval a minimum of 5 weeks prior to testing.
- (c) Test certificates confirming all performance test results including vibration and sound measurements and copies of original test data collection forms shall be submitted for S.O.'s approval.
- (d) Variable speed blowers (if applicable) shall be string tested throughout their entire operating speed range using the motor and inverter to be installed for the Works.
- (e) Air flow rate shall be measured on the discharge side of the blower. Discharge pressure (delivered pressure) shall be measured downstream of the check valve. Input power is the sum of the input electrical power to the blower system (package).
- (f) The blower driver balance test to verify the performance and wire power consumption (including all ancillary power consumption related to running the blower i.e. cooling units, PLC's and related equipment) shall be witnessed by the Board at the Board's discretion. The power rating of ancillary equipment shall be added to the tested power consumption if not included in the performance test.
- (g) The test shall be carried out in a certified test facility. All test equipment shall be calibrated and certified by an independent test agency no more than 12 months prior to the test date. Certificates shall show the stability of calibration over a period of at least 1 year per ISO 9001. In case harmonic currents or currents of high frequencies can be foreseen, the power shall be measured by a power analyser with a band width of 0.1 – 1 MHz and the current transformer shall be able to measure not less than 500 kHz.
- (h) To verify the blower performance over the whole operation range and operation mode based on the control philosophy, the following operation parameters shall be the basis for the test:

<u>Operational levels</u>	<u>Allowed tolerances</u>
• 100% Operation	• Flow $\pm 0\%$
• 80% Operation	• Pressure $\pm 0\%$
• 60% Operation	• Power $\pm 4\%$
• 45% Operation	

- (i) The blowers shall be installed, tested and commissioned at site in accordance with the manufacturer recommendations, relevant specifications and as described herein:
 - (i) Factory trained engineers of the manufacturer, who have complete knowledge and experience in the proper installation, startup and operation of the equipment, shall inspect the final installation and assist the Contractor with all field testing and commissioning of the equipment.

- (ii) Noise Test: Noise testing shall be conducted during system testing. The manufacturer shall verify operations are within specified tolerances.
- (j) For HST blowers, the ASME PTC10 code shall be supplemented with the following requirements:
 - (i) Test shall not be limited to only the blower core but shall include the blower package, i.e. turbo blower, motor, VFD, local control panel, harmonics filter, cooling systems for the control and VFD, blow-off valve, discharge diffuser cone, discharge check valve, discharge expansion joint and enclosure;
 - (ii) Inclusion of inlet filter loss in both calculation of the bid guarantee and actual test results;
 - (iii) Allowable tolerance on volume flow, isentropic head and Wire-to-Air Power to be $\pm 1\%$, $\pm 1\%$ and 1% respectively; and
 - (iv) Minimum of 5 test points.
- (k) For HST blowers, the on-site testing and commissioning shall be supplemented with the following Harmonic Distortion Tests. Harmonics level shall meet the requirement specified in IEEE 519-2014:
 - (i) The manufacturer shall conduct a harmonic distortion tests at site as described here in;
 - (ii) With each blower, as well as combinations of each blower up to the full operating load, measure current harmonic distortion at the PCC for all harmonics up to 35th harmonic;
 - (iii) Show that the percent current harmonic distortion is below the limits as specified in IEEE 519-2014;
 - (iv) Measure total voltage distortion at the MCC with two new blowers operating at full load;
 - (v) Measured results should match with the approved calculations submitted by the blower manufacturer;
 - (vi) Provide distortion analyser, current, and potential transformers required for the test set up;
 - (vii) Submit a test plan for review and approval prior to implementing the actual test. An approved test plan is mandatory before conducting a test;
 - (viii) Provide at least 2 weeks' notice before conducting test;
 - (ix) Submit site test reports to S.O. for approval.

G4.10.9 Control Requirements

- (a) General (All Blowers)
 - (i) The general guide line for the control of the air blower is described in this section and the details of control shall be fully in accordance with the ICA and PCN specifications.

- 1) The control system shall be designed so that the duty allocation of the blowers may be selected in any sequence. This selection shall be able to be made at both the Blower System Master Control Panel and at the SCADA system;
- 2) Each blower system shall be fully controlled by the Blower Master Control Panel with monitoring, alarms, key setpoint adjustments, and individual blower mode selection and start-stop selection, available via a touch-screen HMI on the MCP and also possible from the plant SCADA system;
- 3) The blower shall be designed to be fail-safe, such that failure of any essential components/systems shall generate an alarm and, if necessary, shut-down the blower;
- 4) The blower shall start and stop automatically as required by its Control System;
- 5) During starting, the blower shall not dwell at or near any critical speed, but shall rapidly accelerate through it;
- 6) Normal Operation;
 - Manual Mode
Starting and stopping of the blower shall be performed manually with 'start' and 'stop' push-buttons on the LCP.
 - Off Mode
The blower shall coast to a stop under its own inertia. This position shall also be used to clear a fault and allow restarting of the blower after it has shut down for any fault condition.
 - Automatic Mode
The blower shall start and stop automatically as required by its Control System.
- 7) The blower shall automatically shut-down (or starting shall be inhibited) under the following fault/alarm conditions. Re-starting of the blower shall not be possible unless the fault/alarm is manually reset at the LCP:
 - Motor starter fault (overload, earth fault etc.) (all drives);
 - High drive motor insulation temperature (thermistor trip);
 - High outlet air pressure;
 - High outlet air temperature;
 - Low oil pressure (if appropriate);
 - High oil temperature (if appropriate); and
 - High inlet filter differential pressure.
- 8) Operation of an emergency stop pushbutton on the LCP (unless otherwise specified) shall cause the blower and all auxiliary plant to stop in both Hand and Automatic modes; and

- 9) No alarm or shutdown shall isolate the LCP display, if an LCP is provided.
- (b) Master Control Panel (all types of blowers)
 - (i) The MCP shall bring blowers online and offline, increase/decrease online blower capacity resulting in a gradual increase/decrease of air throughout the entire range of one to multiple blowers online. In the event of a blower failure, the next blower in the preselected start sequence shall come online. Provide control scheme to operate with a minimum number of blowers online while maintaining maximum system efficiency.
 - (ii) On return of power, following a loss of plant power, the blowers shall not restart until the SYSTEM START is initiated by the plant control system or by operator from the MCP. On receiving the RUN command, the MCP shall initiate a sequenced re-start of the blowers.
 - (iii) The touch screen HMI for the MCP shall be configured to display multiple graphic screens for displaying operating variables, valve positions, and other relevant data. Operating screens shall include as a minimum:
 - 1) Screen for start sequence selection of blowers, and blower SYSTEM START initiation. The blowers shall start in the selected sequence when SYSTEM START is initiated from the MCP or by the plant control system;
 - 2) Pressure and flow setpoint adjustments for operation in LOCAL mode;
 - 3) Displays and controls shall be provided to monitor all process variables;
 - 4) Monitor and modify all process related setpoints, as required;
 - 5) Display all alarm conditions within entire blower system; and
 - 6) Control of the blowers and air flow as a manual function.
 - (iv) As a minimum, the MCP shall have status indicators for each blower as follows:
 - 1) Blower in remote;
 - 2) Blower ready for start;
 - 3) Blower running;
 - 4) Common alarm;
 - 5) Increase air flow; and
 - 6) Decrease air flow.
 - (v) MCP Interfaces with Plant Control System (SCADA):
 - 1) All control, monitoring, and alarm signals available to the MCP shall be made available to the plant SCADA via the Process Automation Network. Data shall be organized within contiguous memory blocks in the MCP PLC to simplify coordination and interface with the plant SCADA.
 - 2) MCP Control Interfaces with the plant SCADA shall include but not be limited to the following:
 - Blowoff valve position;

- Discharge header pressure;
 - Discharge air flow (Calculated);
 - Air flow setpoint;
 - MCP ON status;
 - MCP FAIL status;
 - MCP IN REMOTE;
 - MCP READY;
 - Blowoff valve position OPEN;
 - Blowoff valve position CLOSED;
 - Header pressure HIGH alarm;
 - Header pressure LOW alarm; and
 - MCP SHUTDOWN command.
- (vi) The MCP shall provide the following selectable Blower System Control schemes:
- 1) Header Pressure Control: In normal operation the plant control system will provide a pressure set point to the blower master PLC (MCP). The master PLC shall generate commands to the LCPs to start/stop blowers and adjust blower capacity to maintain the air header at the pressure set point. The MCP shall provide air header pressure control with any or all of the blowers in service and shall provide control scheme to operate with a minimum number of blowers online while maintaining a maximum system efficiency. The pressure set point shall be adjustable from the MCP when set to LOCAL mode, or from the plant control system when set to REMOTE mode. In the REMOTE mode, the setpoint adjustable range shall be subject to limits set within the MCP.
 - 2) Header Flow Control: In normal operation the plant control system will provide flow set points to the MCP. The MCP shall generate commands to the blower LCPs to start/stop blowers and adjust blower capacity to deliver the required flow. The MCP shall provide air header flow control with one or more blowers in service and shall provide a control scheme to operate with a minimum number of blowers online while maintaining a maximum system efficiency. The flow set point shall be adjustable from the MCP when set to LOCAL mode, or from the plant control system when set to REMOTE.
 - 3) If a Process Blower fails to start or fails during operation, a standby blower shall be started.
 - 4) Header Blow-Off Valve Control: The blowoff control valve on the process header shall be used during low flow conditions to prevent blowers from shutting down due to surge conditions, and may be used during sequencing of blowers. The MCP shall periodically check the status of the machine operation and attempt to close the blow-off valve with each review cycle.

5) In no case shall any two blowers start simultaneously.

(c) Centrifugal Blowers

- (i) The blowers shall start under an automatic sequence initiated by the local start signal or the remote start signal from the MCP when in REMOTE control. Upon signal to start, contacts shall signal that the inlet guide vanes and variable diffuser vanes are at minimum, the bypass valve is opened and the discharge valve is closed. The blowers shall start under an automatic sequence initiated by the local start signal or the remote start signal from the MCP when in REMOTE control. Upon signal to start, contacts shall signal that the inlet guide vanes and variable diffuser vanes are at minimum, the bypass valve is opened and the discharge valve is closed. All vanes and valves shall be equipped with limit switches on both the open and closed position to indicate position. If components are not properly positioned, they shall move to their respective start positions automatically via the PLC logic.
- (ii) The oil pre-lubrication system shall energize and run for a minimum 2-minute pre-lubrication time. Once all start/safety functions are confirmed, the blower motor shall be started. A feedback signal from the MCC (by others) shall confirm the main drive motor starter being energized. When the blower reaches operating speed, as determined by the start sequence, the contacts shall operate to open inlet guide vanes, open discharge valve, close blow-off (bypass) valve and release control of the inlet guide vanes and variable diffuser vanes to local/remote control. Interlocks shall be provided to prevent blower operation after the delay time during start-up, if the components are not correctly positioned. Provide sequence fail alarm and shut-down if any portion of the start, run, or stop sequence is not properly executed.
- (iii) The surge detection system shall sense unbalanced/surge conditions by use of flow, pressure or temperature sensing devices. Detection of surge conditions shall shut-down the blower.
- (iv) There shall be four means of shutting down the blower:
 - 1) Normal Stop: Initiated by pushing the local stop button or,
 - 2) Remote stop from the MCP. Machine normally stops such that no surging occurs.
 - 3) Soft Stop Initiated by: High Oil Temperature; High Inlet Air Temperature (if required by manufacturer); High Motor Temperature (MCC Alarm) or Engine Malfunction; Discharge Valve is not fully open within 2 minutes after receiving feedback signal from MCC; Blower Bypass Valve has not closed within 5 minutes after receiving feedback signal from MCC; or Surge.
 - 4) Emergency Stop Initiated by: Pushing emergency stop button; LOW-LOW oil pressure; No feedback signal from MCC during start sequence; Loss of feedback signal from MCC during normal operation; Bearing Temperature at Danger; or LCP PLC failure.
- (v) The shaft rotation switch shall be active only when there is no main motor running feedback present at the LCP. The purpose of both of these sensors is to detect reverse air flow through the compressor.

- (d) High Speed Turbo Blower
 - (i) The blowers shall start under an automatic sequence initiated by the local start signal at the LCP when in LOCAL control, or the remote start signal from the MCP when in REMOTE control. All blowoff valves shall be equipped with limit switches on both the open and closed position and remote status if the valve is provided with a local override to indicate position. If components are not properly positioned, they shall move to their respective start positions automatically.
 - (ii) There shall be three means of shutting down the blower:
 - 1) Normal Stop: Initiated by pushing the local stop button or remote stop from the MCP. Machine normally stops such that no surging occurs;
 - 2) Soft Stop Initiated by surge; and
 - 3) Emergency Stop Initiated by: Pushing EMERGENCY STOP button; or LCP PLC failure.
 - (iii) The surge detection system shall bypass air through the blowoff valve or shut down the blower.
 - (iv) The blower machine control and protection system shall include the following instrumentation, as a minimum:
 - 1) Blower inlet air temperature transmitter;
 - 2) Blower discharge air temperature transmitter;
 - 3) Blower inlet air pressure transmitter;
 - 4) Blower discharge air pressure transmitter;
 - 5) Inlet air filter differential pressure Transmitter;
 - 6) The temperature monitoring system shall monitor and display actual winding temperatures at the LCP. A high temperature (as determined by the blower manufacturer) shuts down the blower and gives an alarm. The PLC shall receive the alarm and the operator interface shall graphically display the temperature signals. The alarm/shutdown shall be displayed until reset. Provide necessary hardware for direct communication between temperature monitors, PLC, and Operator Interface;
 - 7) Air flow measurement indication (calculated); and
 - 8) Blower speed and indication.

G4.10.10 Associated Plant and Equipment

- (a) Air Pipework
 - (i) Air pipework systems shall be well designed and maintainable. The air flow velocity shall be selected to minimise the high-pressure loss in the system, prevent any adverse pressure drop and any carry over of moisture past the drain legs. The air flow velocity shall not be less than 6 m/s but not exceed 15 m/s. The design and associated calculations shall be endorsed by a P.E. of the relevant discipline and submitted to the S.O. for approval.

- (ii) Metal pipework shall be earthed to discharge static electricity.
- (iii) Drainage points shall be provided and at not less than 30 m intervals. Drainage points shall be formed by equal tees with a down pointing leg fitted preferably where changes of direction occur.
- (iv) Branch take offs shall be from the top of the main. The bottom of any falling pipe shall be drained.
- (v) The pipework system shall be fully hydraulically pressure tested.
- (vi) The pipework system shall be designed to avoid resonance and vibration.
- (vii) Thermal expansion and contraction system shall be included in the pipework system. Sufficient and suitable bellow type expansion joint with supports, guiding pipe and control rod shall be provided to accommodate the pipe movement. The metal bellow and guiding pipe, shall be made of stainless steel 316 as minimum, unless otherwise specified. The details of the pipework arrangement shall be submitted to the S.O. for approval.
- (viii) Pipework up to and including 50 mm N.B. shall be in steel to BS EN 10255 with heavy fittings in malleable iron to BS 143 screwed to BS EN 10226-1. Threaded joints shall be made with a temperature resistant jointing compound. PTFE tape shall not be used.
- (ix) Pipework over 50 mm N.B. shall be in steel to BS EN 10216 grade 360 or 430 with flanged joints drilled to BS EN 1092 PN10 or higher rating where applicable, unless otherwise specified.
- (x) Care shall be taken at points where condensation is likely to form or to accumulate such as following compression or cooling of ambient air or discharge of reactivation air from dryers.
- (xi) All main pipes and branch pipelines shall slope downward at a gradient of at least 1/1200 in the direction of air flow to allow condensation to flow down drip legs and collect at the low points for removal.
- (xii) Appropriate traps and automatic drain valves discharging to approve points shall be installed at the bottom of drip legs. The discharging points shall be subjected to S.O.'s approval.
- (xiii) Branch pipelines tapping points shall be from the crown instead of invert of main pipes.
- (xiv) Thermal insulation and cladding or protection mesh wire guard where fitted as per specified and shall be arranged in a manner which provides access to valves and other components for maintenance purposes and shall be subjected to S.O.'s approval.
- (xv) Pipework which is connected to blowers shall be via flexible connectors, with the pipework immediately downstream of the flexible joint being suitably supported.
- (xvi) Test points shall be provided at the outlet and inlet pipework and shall be proposed and submitted to the S.O. for approval.
- (xvii) Wall penetrations and pipework supports shall be designed to limit the transmission of noise / vibration; and

- (xviii) Equipment foundations, supports and restraints, etc. shall be designed and installed with due regard to all static and /or dynamic forces that will be imposed upon them by the installed equipment, including, as a minimum, forces arising due to surge, thermal effects and vibration.
- (b) Air Flow Control Valves
 - (i) Air flow regulating valves shall be globe type constructed of stainless steel 316C16 to BS EN 10213. Valve spindle shall be stainless steel type 316S16 to PD 970. Gland packing and seating ring shall be PTFE. Valves shall be sized in accordance with BS EN 60534-2-3. Flanged valves shall have integral flanges drilled to BS EN 1092 with pressure designation to match the pipework system. The flow control range of the valves shall be 10:1 or greater.

END OF SECTION